



MD MAHABUBUR RAHMAN

FULL-STACK DEVELOPER & CYBERSECURITY SPECIALIST & ML ENGINEER

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PROFESSIONAL PROFILE

Innovative **Full-Stack Developer**, **Cybersecurity Specialist**, and **Machine Learning Engineer** with expertise spanning **Red Team (Offensive)** and **Blue Team (Defensive)** security operations. Proficient in building scalable, secure applications using **MongoDB, Express, React, Node.js, and FastAPI** with **Secure SDLC** principles. Advanced proficiency in **Machine Learning** and **Data Science**, specializing in **supervised learning, ensemble methods, feature engineering, and predictive modeling**. Experienced in vulnerability assessment, penetration testing, SIEM implementation, and digital forensics. Passionate about architecting **AI-powered security solutions** for next-generation threat detection and mitigation.

TECHNICAL EXPERTISE

Full-Stack Development	MongoDB, Express.js, React, Node.js, FastAPI, JavaScript (ES6+), Tailwind CSS, REST APIs, JWT, Docker.
Machine Learning	XGBoost, CatBoost, LightGBM, Random Forest, Decision Trees, SVM, Logistic Regression, Linear Regression, KNN, Naïve Bayes.
Feature Engineering	TF-IDF, Count Vectorization, PCA, Feature Selection, Dimensionality Reduction, One-Hot Encoding, Standardization, Normalization.
Model Optimization	GridSearchCV, RandomizedSearchCV, Cross-Validation, Hyperparameter Tuning, Ensemble Methods (Bagging, Boosting, Stacking).
Evaluation Metrics	Accuracy, Precision, Recall, F1-Score, ROC-AUC, Confusion Matrix, MSE, R-Squared.
Data Science	Python, NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, EDA, Data Visualization, Statistical Analysis.
Offensive Security	Penetration Testing, AD Exploitation, Privilege Escalation, CVE Exploits, Pivoting, RATs, MITM, Phishing, Social Engineering.
Defensive Security	SOC Operations, SIEM (Splunk, Wazuh), EDR, Snort, Wireshark, YARA, Network Analysis, Threat Intelligence.
Digital Forensics	Autopsy, Malware Analysis, Keylogger Research, Browser Security, Dark Web Research.
Programming	Python, JavaScript, Bash, C/C++, PowerShell, SQL.

KEY PROJECTS

MDefender Pro – AI-Powered Malware Detection & WAF

- Designed an AI-powered security platform for WordPress malware detection and web application protection.
- Integrated **machine learning-based malware detection** with real-time scanning, REST APIs, and automated threat classification.
- Implemented file integrity monitoring, license management, secure authentication, and OWASP Top 10 protection.
- Tech Stack: **Python (FastAPI), React.js, Node.js, MongoDB, Docker, Scikit-learn.**

Machine Learning for Cybersecurity

- Developed and benchmarked classification models including **XGBoost, CatBoost, LightGBM, Ran-**

dom Forest, SVM, and Logistic Regression for malware and phishing detection.

- Applied **TF-IDF vectorization** and feature engineering for malicious URL/text classification.
- Conducted **EDA** using Pandas, NumPy, and Matplotlib; optimized models via **GridSearchCV** and cross-validation.

Secure Full-Stack Web Application

- Built a secure application using **React.js, Node.js, Express.js, and MongoDB** with JWT authentication, Bcrypt hashing, and RBAC.
- Applied OWASP Top 10 security controls and conducted testing using **Burp Suite** and **OWASP ZAP**.

Enterprise Red Team vs. Blue Team Lab

- Deployed an enterprise lab with **Windows Server AD, Wazuh SIEM, and Splunk**.
- Simulated attacks: phishing, privilege escalation, lateral movement, and domain compromise.
- Developed detection rules and conducted forensic investigations using **Autopsy**.

Malware Analysis & Detection Engineering

- Analyzed **RATs, Keyloggers, Web Shells, and Backdoors** in isolated environments.
- Authored **YARA** and **Snort** rules; documented IOCs and mitigation strategies.

EDUCATION

B.Sc. in Computer Science & Engineering

Expected 2026

Shanto Mariam University of Creative Technology, Dhaka

Relevant Coursework: Network Security, Cryptography, Secure SDLC, Digital Forensics, Machine Learning, Data Science, Statistics

CERTIFICATIONS & ACHIEVEMENTS

- 🏆 **PortSwigger:** Web Security Academy Practitioner
- 🏆 **TryHackMe:** Active Directory, Network Pentesting, Linux/Windows Privilege Escalation
- 🏆 **Bug Bounty Hunter:** Active vulnerability researcher
- 🏆 **ML Practitioner:** Scikit-learn, XGBoost, CatBoost, LightGBM, Pandas, NumPy, Matplotlib, Seaborn
- 🏆 **CTF Competitor:** Web, Binary Exploitation, Forensics